Experiment 3: What the Pose Error Rises With

Insight analysis for Chapter 4, Section 4.3

Bolton Athit Davies

Reporting cutoff: 22 October 2026; real-robot section added 2 November 2026, rebuilt 3 November (realv6) with realv7 added the same day 2026; revised 4 November 2026

Purpose of this document. The raw write-up of Experiment 3 (`insights/experiment_3/`) reports what each run scored. This document asks *why*: whether the pose error rises when the robot turns, when a movable object is in the camera's view, or after the tracker loses features, and what the robot is doing at the moment the error starts to grow. Every question is tested within each run, so that the run is its own control, and the answer is stated with the evidence that supports it and the evidence that is missing. Every number is written by `script/exp3_insights.py` into `output/compare/simulation/exp3-insights/`; the tables below are input from there unchanged.

1 Questions and method

Five questions were put to the 114 Experiment 3 runs of both testbeds (the registry of `script/analyse_exp3_timelines.py`: 64 dynamic and 18 static runs in the extreme world, 26 dynamic and 6 static in the realistic world).

- Q1. Does the error grow faster while the robot turns than on straights?
- Q2. Does it grow faster while a movable object is in the camera's view?
- Q3. Is the onset preceded by a loss of tracked features?
- Q4. What is the robot doing at the onset: turning or straight, object in view or not?
- Q5. Is the error that these questions concern the estimator's error at all, or does part of it come from how the estimate is scored?

Within-run contrasts. The error series is the start-anchored planar error of the timeline analysis: the estimate is aligned to the ground truth by a rigid fit over the first 10–40 s, after the initialisation transient and before the ground truth has turned 60°, and the planar distance to the ground truth is followed from there. For Q1 and Q2 the samples of a run are split into two states and the growth rate of the error (metres per second of positive and negative change) is computed in each. States are: *turning*, ground-truth yaw rate above $10\,^\circ\,\mathrm{s}^{-1}$; *object in view*, for the extreme bags the projected simulator truth of Experiment 2 (image area of moving objects in the left camera), for the realistic bags the carriers and carried loads whose poses the bag records on `/world/default/pose/info`, placed within 20 m and $\pm 45^\circ$ of the ground-truth heading with no shelf footprint on the line of sight. Rates are taken only while the estimate is still tracking: after the anchor window, before its first impossible-speed jump and while the error is below 3 m. A contrast needs at least 3 s in each state; across runs the two rates are compared with the sign count and the Wilcoxon signed-rank test. For Q3 the matched-feature count (ORB-SLAM3 map inliers, VINS-Fusion tracked features) is followed around the onset as a percentage of the run's own level before it. For Q4 the state at the onset time is read off. The onset is the one the raw write-up and the timeline tables use: 1 s of sustained excess over three times the static median error, with the ground truth scored as the chapter scores it. The drift contrasts of Q1 and Q2 are instead computed with the ground truth at the IMU point (Insight 1), because at the model origin they measure the lever arm, not the estimator.

Lever arm (Q5). Both estimators report the pose of the IMU: the ORB-SLAM3 wrapper converts the camera pose with the camera-to-body extrinsic before writing it, and VINS-Fusion writes its body state. The ground truth is the model's odometry, which reports the model origin. In the simulated platform the IMU sits 0.338 m ahead of that origin (`models/ackermann_robot/model.sdf`). No scoring step in the project compensates this; Chapter 3 states the requirement only for the real robot's tracker marker. The effect is tested by moving the ground truth forward along its own heading by a lever arm L and re-scoring every tracked unfiltered run for L from 0–0.60 m.

2 Insight 1: the error that appears at turns is the scoring lever arm

Scored as the chapter scores them, the error of the tracked runs grows while the robot turns and holds on straights; scored at the IMU point, that pattern disappears and the whole-run ATE falls by half or more. Tables I.1 and I.2 and figure I.1 carry the sweep.

Table I.1: Lever-arm sweep. Both estimators report the IMU frame, which sits 0.338 m ahead of the model origin that the ground truth reports (`models/ackermann_robot/model.sdf`); no scoring step compensates this. The table re-scores the tracked unfiltered runs with the ground truth moved forward by L along its heading: whole-run aligned planar ATE (median over runs) and the drift rate while turning and on straights (median). If the error that appears at turns were the lever arm, ATE and the turning rate would reach their minimum near $L = 0.34$ m.

Testbed	Estimator	n	$L = 0.00$ m			$L = 0.17$ m			$L = 0.34$ m			$L = 0.50$ m		
			ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.
extreme	VINS-Fusion	9	0.32	2.2	-2.1	0.19	7.6	-0.7	0.10	5.2	0.6	0.14	-0.6	1.4
extreme	ORB-SLAM3	9	0.37	-8.2	2.1	0.28	-1.0	2.2	0.19	-0.9	4.7	0.22	-8.1	3.6
realistic	VINS-Fusion	4	0.37	-0.0	2.8	0.29	3.6	5.5	0.25	1.9	5.7	0.28	7.0	3.4
realistic	ORB-SLAM3	8	0.32	8.7	-1.2	0.19	5.4	-0.1	0.15	0.0	2.0	0.24	-2.8	2.0

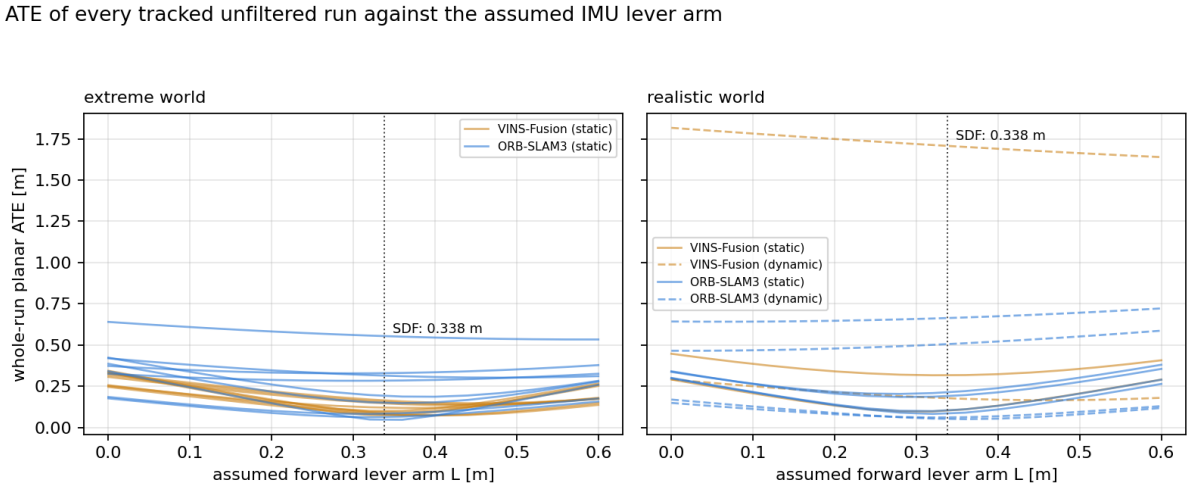


Figure I.1: Whole-run planar ATE of every tracked unfiltered run against the lever arm assumed between the model origin and the IMU. Solid lines are static recordings, dashed the realistic dynamic recordings. The dotted line is the value in the robot model.

The minimum of the ATE lies at the lever arm the robot model specifies. Over the 30 tracked unfiltered runs the ATE-minimising L^* has a median of 0.34 m and an interquartile range of 0.32–0.40 m (table I.2), for both estimators and both testbeds. The three runs that do not follow the pattern are the ones whose error is not a rigid offset: the two ORB-SLAM3 runs on `realistic_002`, whose error is a 0.5–1.0 m step taken in the first 15 s (figure I.4), and the VINS-Fusion run on the same bag, which diverges at its first turn. With $L = 0.34$ m the median static ATE of the extreme world falls from 0.32 m to 0.10 m for

Table I.2: The forward lever arm that minimises each tracked unfiltered run's whole-run planar ATE, from a sweep of L in 0–0.60 m in 0.04 m steps, with the ATE at $L = 0$ (as reported in the chapter) and at the SDF value $L = 0.338$ m. The IMU sits 0.338 m ahead of the model origin in `models/ackermann_robot/model.sdf`.

Testbed	Recording	Estimator	L^* [m]	ATE(0) [m]	ATE(L^*) [m]	ATE(0.338) [m]
extreme	`static_000`	ORB-SLAM3	0.28	0.37	0.33	0.33
extreme	`static_000`	ORB-SLAM3	0.48	0.42	0.30	0.31
extreme	`static_000`	ORB-SLAM3	0.60	0.64	0.53	0.55
extreme	`static_000`	VINS-Fusion	0.44	0.31	0.13	0.15
extreme	`static_000`	VINS-Fusion	0.44	0.32	0.14	0.16
extreme	`static_000`	VINS-Fusion	0.44	0.33	0.15	0.16
extreme	`static_001`	ORB-SLAM3	0.32	0.18	0.06	0.06
extreme	`static_001`	ORB-SLAM3	0.32	0.33	0.28	0.28
extreme	`static_001`	ORB-SLAM3	0.32	0.19	0.08	0.08
extreme	`static_001`	VINS-Fusion	0.40	0.26	0.07	0.08
extreme	`static_001`	VINS-Fusion	0.40	0.25	0.07	0.08
extreme	`static_001`	VINS-Fusion	0.36	0.25	0.12	0.12
extreme	`static_002`	ORB-SLAM3	0.36	0.39	0.15	0.15
extreme	`static_002`	ORB-SLAM3	0.40	0.42	0.19	0.19
extreme	`static_002`	ORB-SLAM3	0.36	0.34	0.05	0.05
extreme	`static_002`	VINS-Fusion	0.32	0.33	0.10	0.10
extreme	`static_002`	VINS-Fusion	0.36	0.34	0.09	0.09
extreme	`static_002`	VINS-Fusion	0.32	0.33	0.07	0.08
realistic	`realistic_000`	ORB-SLAM3	0.28	0.34	0.20	0.21
realistic	`realistic_000`	ORB-SLAM3	0.28	0.34	0.18	0.19
realistic	`realistic_000`	VINS-Fusion	0.32	0.45	0.32	0.32
realistic	`realistic_001`	ORB-SLAM3	0.32	0.30	0.10	0.10
realistic	`realistic_001`	ORB-SLAM3	0.32	0.30	0.08	0.09
realistic	`realistic_001`	VINS-Fusion	0.32	0.29	0.10	0.10
realistic	`realistic_002`	ORB-SLAM3	0.08	0.64	0.64	0.66
realistic	`realistic_002`	ORB-SLAM3	0.00	0.47	0.47	0.51
realistic	`realistic_002`	VINS-Fusion	0.60	1.82	1.64	1.71
realistic	`realistic_003`	ORB-SLAM3	0.32	0.15	0.06	0.06
realistic	`realistic_003`	ORB-SLAM3	0.36	0.17	0.05	0.05
realistic	`realistic_003`	VINS-Fusion	0.48	0.29	0.17	0.18
median L^* = 0.34 m, IQR 0.32–0.40 m over 30 runs						

VINS-Fusion and from 0.37 m to 0.19 m for ORB-SLAM3; in the realistic world from 0.37 m to 0.25 m and from 0.32 m to 0.15 m (table I.1).

The turning effect goes with it. At $L = 0$ ORB-SLAM3's error in the realistic world grows at a median 8.7 mm s^{-1} while turning and shrinks on straights, in eight of eight unfiltered runs; at $L = 0.34$ m the turning rate is 0.0 mm s^{-1}. This is the signature of a rigid offset between the scored point and the reference point: the offset vector rotates with the robot, so the apparent error changes only while the heading changes.

The sweep fits one parameter and could, on its own, absorb some estimator error. What makes it a finding rather than a fit is that the minimum coincides with an independently specified geometric constant, in 30 runs, across two estimators that share no code. The residual ATE at L^* is the estimator's.

[Consequence beyond this experiment] Every simulation ATE in Chapter 4 that aligns an estimate to the model-origin ground truth carries this offset: the static floor of Experiment 3 and the visual estimators of Experiment 1. The wheel and IMU baselines of Experiment 1 are referenced to the same origin and are not affected in the same way, so the estimator ranking of that experiment has to be re-scored before it is quoted. The correction is one line: move the ground truth to the IMU point before alignment. It has since been applied in the scoring scripts (`vio_metrics.load_gt`, 22 October 2026) and every simulation table and figure of Experiments 1 and 3 has been regenerated; the chapter prose that quotes the old values is pending.

3 Insight 2: after the correction, turning does not raise the error

With the ground truth at the IMU point, neither estimator's error grows faster while turning than on straights, in either testbed, and the heading error does not either. Tables I.3 and I.4 give the contrasts; figure I.2 plots each run.

Table I.3: Within-run drift contrasts, extreme world. Each run is its own control: the growth rate of the start-anchored error is computed separately in the two states and the two rates are compared across runs (median of each, number of runs where the first exceeds the second, Wilcoxon signed-rank p). "Turning": ground-truth yaw rate above 10 ° s^{-1}. "Object in view": detector boxes present on the frame (borrowed from the class-filtered run on the same bag for unfiltered runs). Rates are taken only while the estimate is still tracking (before the first jump and below 3 m error). Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	n	turning	straight	$n_>$	p
			[mm s^{-1}]			
static	VINS-Fusion	9	5.2	0.6	7	0.10
static	ORB-SLAM3	9	-0.8	4.7	1	0.10
dynamic	VINS-Fusion	2	46.4	169.9	0	—
dynamic	VINS-Fusion+YOLO	2	94.0	180.6	0	—
Heading error, same contrast [m° s^{-1}]						
static	VINS-Fusion	9	9.0	9.1	4	0.73
static	ORB-SLAM3	9	-5.5	4.6	3	0.65
dynamic	VINS-Fusion	2	162.5	169.7	1	—
dynamic	VINS-Fusion+YOLO	2	94.6	103.7	0	—
Object in view vs none, dynamic recordings [mm s^{-1}]						
			in view	none		

The static recordings of the extreme world are the only group with enough runs for a test, and the two estimators point opposite ways there: VINS-Fusion is faster while turning in seven of nine runs and ORB-SLAM3 in one of nine, neither at $p < 0.05$. In the realistic world the sign is mixed in every group. The heading-error contrast is flat throughout. On these data the answer to Q1 is no.

Table I.4: Within-run drift contrasts, realistic world. Each run is its own control: the growth rate of the start-anchored error is computed separately in the two states and the two rates are compared across runs (median of each, number of runs where the first exceeds the second, Wilcoxon signed-rank p). "Turning": ground-truth yaw rate above $10\,^\circ\,s^{-1}$. "Object in view": detector boxes present on the frame (borrowed from the class-filtered run on the same bag for unfiltered runs). Rates are taken only while the estimate is still tracking (before the first jump and below 3 m error). Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	n	turning	straight	$n_>$	p
			[mm s^{-1}]			
static	VINS-Fusion	2	3.9	7.5	1	—
static	ORB-SLAM3	4	0.1	1.6	1	—
dynamic	VINS-Fusion	1	1.8	4.8	0	—
dynamic	ORB-SLAM3	4	-1.3	2.1	2	—
dynamic	VINS-Fusion+YOLO	3	0.1	5.5	1	—
dynamic	ORB-SLAM3+YOLO	9	9.1	4.3	2	0.20
Heading error, same contrast [m° s^{-1}]						
static	VINS-Fusion	2	4.1	3.2	1	—
static	ORB-SLAM3	4	5.6	-6.1	3	—
dynamic	VINS-Fusion	1	7.0	9.4	0	—
dynamic	ORB-SLAM3	4	-15.0	5.8	0	—
dynamic	VINS-Fusion+YOLO	3	0.6	0.1	1	—
dynamic	ORB-SLAM3+YOLO	9	-5.4	3.7	3	0.30
Object in view vs none, dynamic recordings [mm s^{-1}]						
			in view	none		
dynamic	VINS-Fusion+YOLO	1	1.0	4.2	0	—
dynamic	ORB-SLAM3+YOLO	3	14.4	7.0	1	—

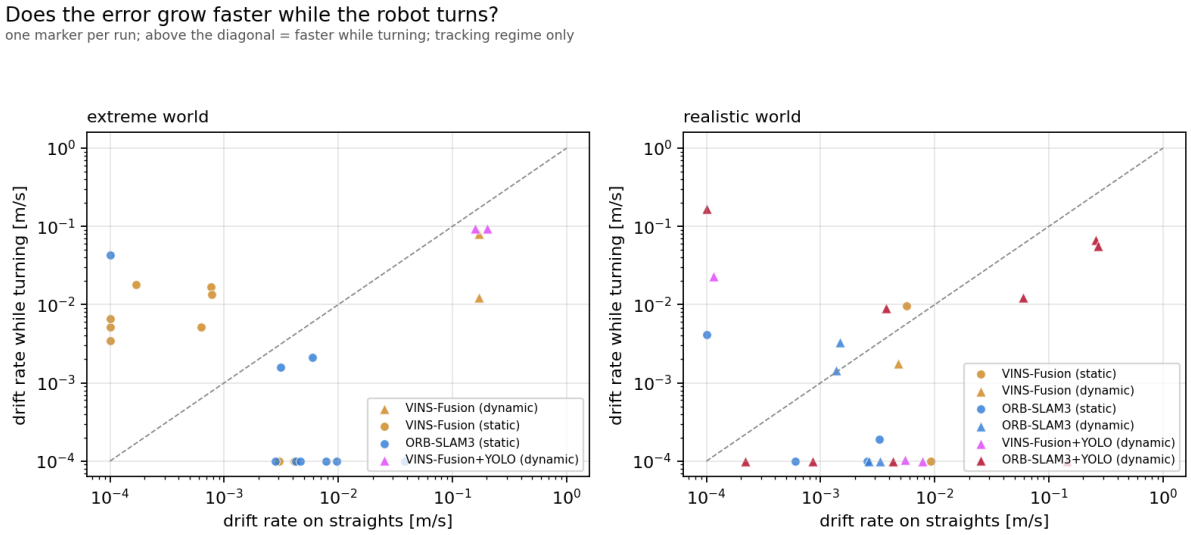


Figure I.2: Drift rate while turning against drift rate on straights, one marker per run, ground truth at the IMU point. Negative rates are drawn at the axis floor. A run above the diagonal drifts faster while turning.

4 Insight 3: in the extreme world the failure is immediate and is not modulated by how much moving object is in view

Every dynamic run of the extreme world passes its onset threshold on the first aisle, almost never while turning, and the moment-to-moment amount of moving object in the image does not predict when ORB-SLAM3 loses tracking or when either estimate jumps. Tables I.5 and I.6 and figure I.3 carry the evidence.

Table I.5: What the robot was doing at the error onset, extreme world, per estimator and condition. "Turning" and "object in view" as in the contrast table; "features" is the median tracked-feature level in the 5 s before onset as a percentage of the run's own level before that. Runs without an onset are counted separately. Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	runs	no onset	turning [of onsets]	object in view [of onsets]	features [% of own]
static	VINS-Fusion	9	9	0/0	—	—
static	ORB-SLAM3	9	6	1/3	—	100
dynamic	VINS-Fusion	12	0	4/12	9/9	97
dynamic	ORB-SLAM3	12	0	0/12	6/6	—
dynamic	VINS-Fusion+YOLO	20	0	3/20	17/17	87
dynamic	ORB-SLAM3+YOLO	20	0	8/20	16/16	89

Table I.6: Extreme world, dynamic recordings: tracking events against the *true* moving-object area in view, from the projected simulator truth of Experiment 2 (`output/yolo_eval/dyn_*/yolo_eval.csv`). Within each run the frames are split into terciles of moving-object area; the table gives the median over runs of the share of frames in which ORB-SLAM3 was not in the OK state, the rate of impossible-speed jumps of the estimate, and the Spearman correlation between matched features (ORB-SLAM3 map inliers, VINS-Fusion tracked features) and moving-object area. $n_>$ counts runs where the top tercile exceeds the bottom; p is the Wilcoxon signed-rank test over runs.

Estimator	n	area med. [%]	top tercile	bottom tercile	$n_>$	p
Share of frames not tracking OK						
ORB-SLAM3	12	5.7	0.195	0.147	5	0.79
Share of frames not tracking OK						
ORB-SLAM3+YOLO	20	5.5	0.132	0.084	10	0.39
Jumps of the estimate per second						
VINS-Fusion	12	5.5	21.65	20.94	7	0.03
ORB-SLAM3	12	5.7	7.04	5.34	6	0.97
VINS-Fusion+YOLO	19	5.4	20.62	18.18	12	0.10
ORB-SLAM3+YOLO	20	5.5	0.70	1.22	10	0.52
Spearman ρ, matched features against moving-object area (median over runs; runs with $\rho < 0$)						
VINS-Fusion	12		ρ = 0.19		3	
ORB-SLAM3	12		ρ = -0.11		11	
VINS-Fusion+YOLO	19		ρ = 0.18		7	
ORB-SLAM3+YOLO	20		ρ = -0.11		14	

Moving objects are in view for 74–91 % of the frames of these four bags (projected truth of Experiment 2), so a within-bag "no object" baseline does not exist; the comparison has to be between more and less moving area. Split into terciles of moving area within each run, the share of frames in which ORB-SLAM3 is not tracking OK is not higher in the top tercile than in the bottom one (table I.6), nor is the jump rate of either estimate. The matched-feature count falls weakly with moving area (median Spearman ρ about -0.1 for ORB-SLAM3, in eleven of twelve runs).

The onset is on a straight. Of the 64 dynamic runs, the robot was turning at onset in fifteen (table I.5); the first heading change of every route comes after the first aisle, and the error has passed threshold before

What precedes the error onset

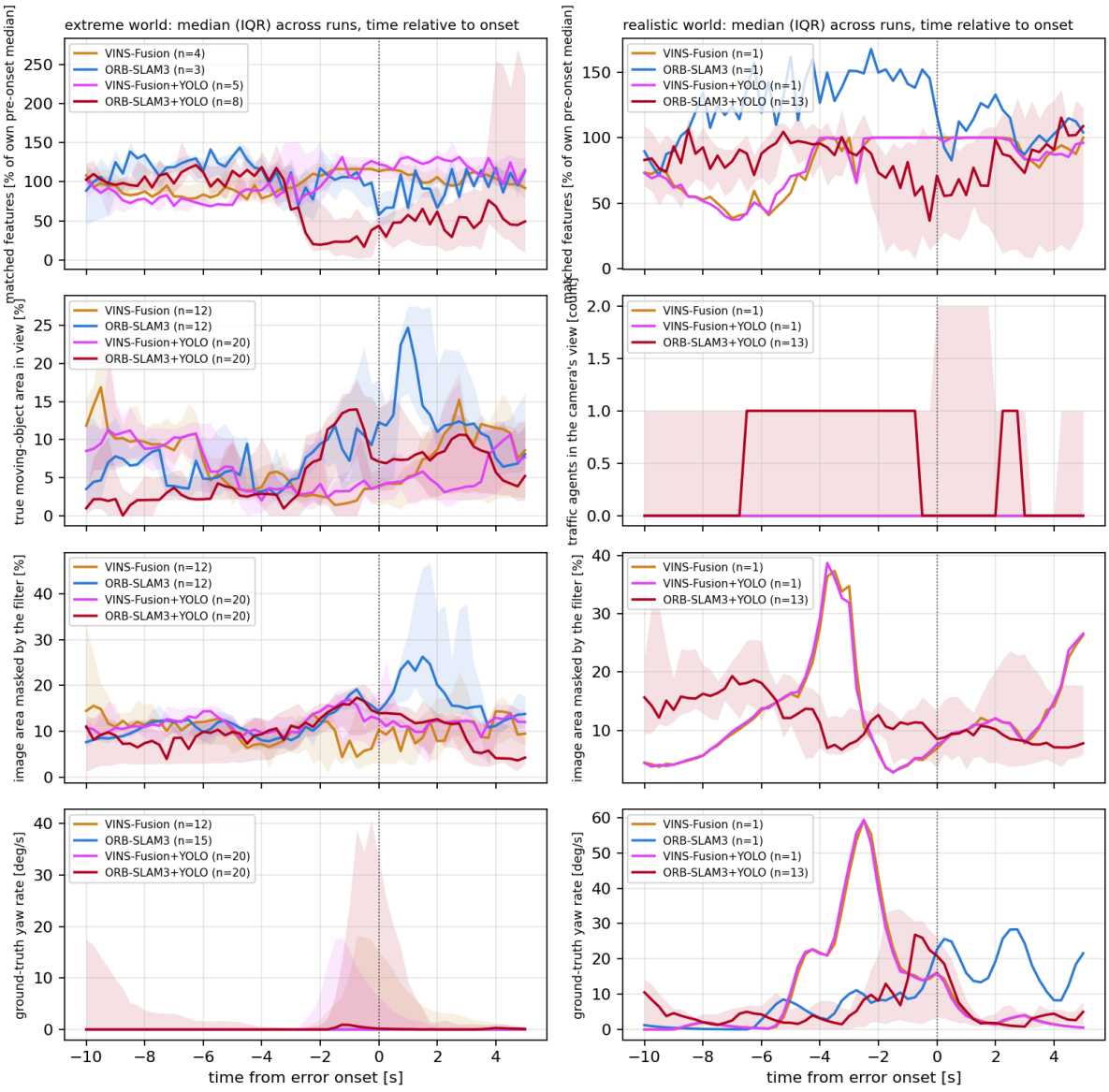


Figure I.3: What precedes the onset, at the chapter's onset definition: median over runs (shaded, interquartile range) of the matched-feature count relative to the run's own pre-onset level, the true moving-object area in view (extreme world) or the number of traffic agents in view (realistic world), the image area the filter masked, and the ground-truth yaw rate, against time from onset.

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The two filtered estimators reach their onset differently. For ORB-SLAM3 with the class mask the matched features collapse to about one fifth of the run's own level 2.5 s before the onset (figure I.3, top left): the mask removes the features the tracker was using, and the error follows. VINS-Fusion shows no feature loss before its onset, filtered or not; its estimate does not drift, it breaks, with a jump rate of about 20 per second once it has failed. What breaks it is not identified by these data; the hypothesis that wrongly tracked moving points corrupt the visual-inertial initialisation remains untested.

5 Insight 4: in the realistic world the traffic in view does not measurably change the drift, and the class mask costs features when traffic is in view

With the carriers' true positions from the bag, the drift of the tracked realistic runs is not consistently faster while a carrier is in the camera's view, and ORB-SLAM3 with the class mask has fewer matched features during those moments without a drift penalty. Table I.7 and figure I.4 carry the evidence; the per-run values are in `output/compare/simulation/exp3-insights/realistic_runs_lever0p338.tex`.

Table I.7: Realistic world, dynamic recordings: within-run contrasts between the moments a traffic agent (carrier or person) is in the camera's view and the moments none is. Visibility is computed from the bag's own model poses (`/world/default/pose/info`): within 20 m, within ±45° of the ground-truth heading, and with no shelf footprint on the line of sight. Drift rate while tracking (before the first jump, below 3 m); share of frames ORB-SLAM3 was not tracking OK; median matched features (ORB-SLAM3 map inliers, VINS-Fusion tracked features); jumps per second. $n_>$ counts runs where the in-view value exceeds the other; p is the Wilcoxon signed-rank test.

Quantity	Estimator	n	in view	none	$n_>$	p
drift rate [mm s^{-1}]	VINS-Fusion	1	2.0	5.7	0	—
	ORB-SLAM3	4	-0.2	1.5	1	—
	VINS-Fusion+YOLO	3	2.4	9.1	0	—
	ORB-SLAM3+YOLO	10	42.0	8.2	7	0.08
not tracking OK [share]	ORB-SLAM3	4	0.000	0.000	0	—
	ORB-SLAM3+YOLO	16	0.003	0.029	5	0.53
matched features	VINS-Fusion	2	150	150	0	—
	ORB-SLAM3	4	431	453	2	—
	VINS-Fusion+YOLO	4	150	150	0	—
	ORB-SLAM3+YOLO	16	350	370	5	0.05
jumps [s^{-1}]	VINS-Fusion	2	0.00	0.00	0	—
	ORB-SLAM3	4	0.00	0.01	0	—
	VINS-Fusion+YOLO	4	0.00	0.00	0	—
	ORB-SLAM3+YOLO	16	0.05	0.07	7	0.82

Traffic was in view for a median 37 % of the tracked time.

Traffic is in view for a median 37 % of the tracked time, so both states are well populated; the groups of runs are not. Two unfiltered VINS-Fusion runs and four ORB-SLAM3 runs carry the unfiltered answer, and in them the drift is, if anything, slower while a carrier is in view (median −0.2 mm s^{-1} against 1.5 mm s^{-1} for ORB-SLAM3, one run of four faster in view). The filtered ORB-SLAM3 group is the only one large enough to test: there the matched-feature count is lower while traffic is in view in eleven of sixteen runs ($p = 0.05$) and the drift rate is higher in seven of ten ($p = 0.08$), with no change in the tracking-loss share. Taken together these suggest, and do not establish, that the class mask costs features when a carrier is in view and that the cost is paid in drift; the unfiltered estimators show no such cost.

The one large degradation of the realistic world is not a traffic encounter. The VINS-Fusion error on `realistic_002`, 1.8 m whole-run ATE, is a single 3 m step taken between 29 s and 33 s, at the end of

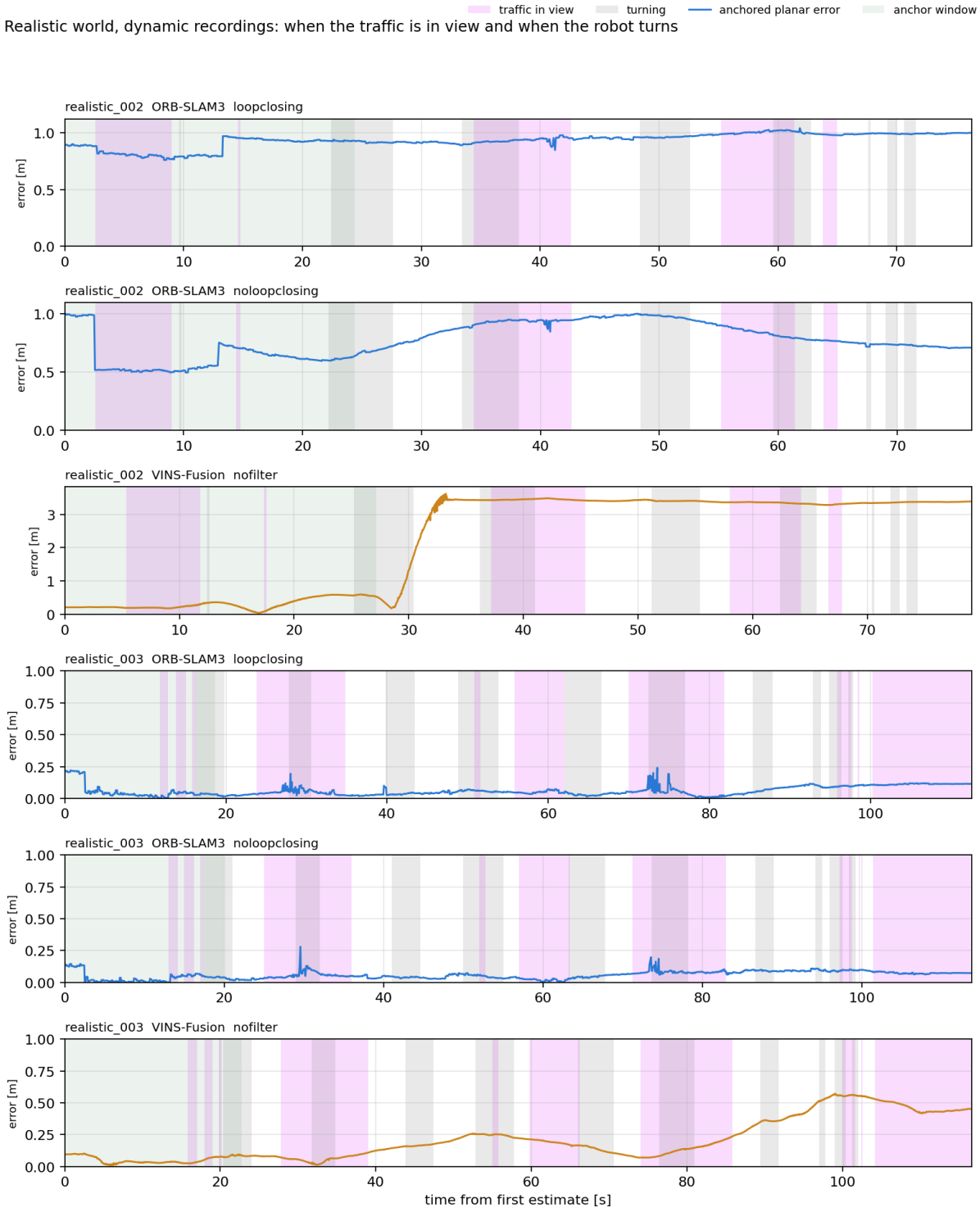


Figure I.4: Realistic dynamic recordings, unfiltered runs: anchored planar error (ground truth at the IMU point) with the moments a carrier or carried load is in the camera's view (violet) and the moments the robot turns (grey). The green band is the alignment window.

the first heading change and with no carrier in the camera's view (figure I.4, third panel); the carrier that passed at 5–12 s left no trace. The same bag gives both ORB-SLAM3 runs their 0.5–1.0 m step at 2.5 s and 13 s, the moments that same carrier enters and leaves the view. Whether the step is the estimator's or the ground truth's is not settled by these data.

[Coverage of the traffic measurement] Walking people are gz-sim actors and do not appear on the pose topic, so their visibility is unmeasured; occlusion is tested against shelf footprints only; and the carried loads move only while carried. The in-view state is therefore a lower bound on the moments something moving was in the image.

6 Insight 5: real robot (`dataset_realv6_000`): ORB-SLAM3 tracks the figure-of-eight to 0.5 m with the GPU front end, loses a factor of two with the CPU front end, and VINS-Fusion has not yet had a valid run

The first real recording with a usable optical-tracker reference since the camera remount gives ORB-SLAM3 a provisional planar ATE of 0.46 and 0.52 m in the two runs with the GPU feature extractor and 0.62 and 1.03 m in the two with the CPU extractor, which drops two thirds of the frames at real-time replay; no run loses tracking after its initialisation resets. VINS-Fusion's run resets 217 times on gaps in its IMU stream and stops after 19 s. Every number rests on a tracker export that fails the scorer's validity rules and on one spurious sample whose removal is documented below. Tables I.8 to I.11 and figures I.5 to I.8 carry the evidence.

The recording. `dataset_realv6_000` was recorded on 31 October 2026 on the Gensurv floor: 220 s, 6 601 stereo pairs, the RealSense IMU at 200 Hz, wheel-speed feedback, and a figure-of-eight route of about 27 m at a median 0.15 m s^{-1}, no scripted moving objects.

The runs. Five runs made on the afternoon of 3 November 2026 are reported, the only ones whose run directories still exist: ORB-SLAM3 stereo-inertial with the GPU or the CPU feature extractor and with loop closing on or off, and VINS-Fusion once; all at replay rate 1.0, all without a detector, so unfiltered. ORB-SLAM3 used `config/wil/stereo_imu.yaml` with the repaired library (fork patches 11–14b, node and library built alike), the 3 November stereo extrinsic and `IMU.NoiseGyro` 0.1 with `IMU.NoiseAcc` 1.0, ten times the values that had been copied from the VINS file; VINS-Fusion used the same file's gravity norm of 9.2642 from an August recording. Earlier runs on this bag (2 November, and the morning of 3 November) were overwritten by these re-runs; their scorings remain under `output/compare/real/realv6*` as a record of the search described in section 6.1 but cannot be reproduced and carry no number here. No filtered run exists, so the filter hypothesis remains untested on the real robot.

Reporting rules. Scoring is `script/eval_real_gt.py`: rigid SE(2) alignment with scale fixed, lever arm from the tracker marker to the IMU unmeasured and set to zero, every number *provisional*. The bag was not on the machine when these runs were scored, so the time offset is pinned to the value the wheel-speed cross-correlation found on 2 November (–4.14 s on the cleaned export, 4.38 s on the raw one) and the bag's first-message time and duration are taken from that day's manifest; the new manifests record the substitution. The scorer's validity rules were fixed before the data were read; the tracker export fails two of them (gaps above 1 s, a step above 0.5 m), so the tool labels every run *failed (ground truth invalid)* and the numbers are reported as diagnostics, not as a result-slot headline. Outputs: `output/compare/real/realv6_1103pm_*/` and `output/compare/real/realv6-insights/`.

One tracker sample decided the headline. The export runs at 1.7 Hz, not the 7 Hz of the earlier recordings, with 66 gaps longer than 1 s, and it contains a single sample 5.2 m off the path. That sample

Table I.8: The same ORB-SLAM3 run (GPU, LC on) scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). The time offsets are the values the wheel-speed cross-correlation found on each export on 2 November (peak correlation 0.10 raw, 0.61 cleaned), applied here as pinned values. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz).

Truth	GT path [m]	offset [s]	ATE [m]	median [m]	path ratio
as delivered	36.8	4.38	1.174	0.851	0.65
one sample removed	27.1	-4.14	0.519	0.437	0.85

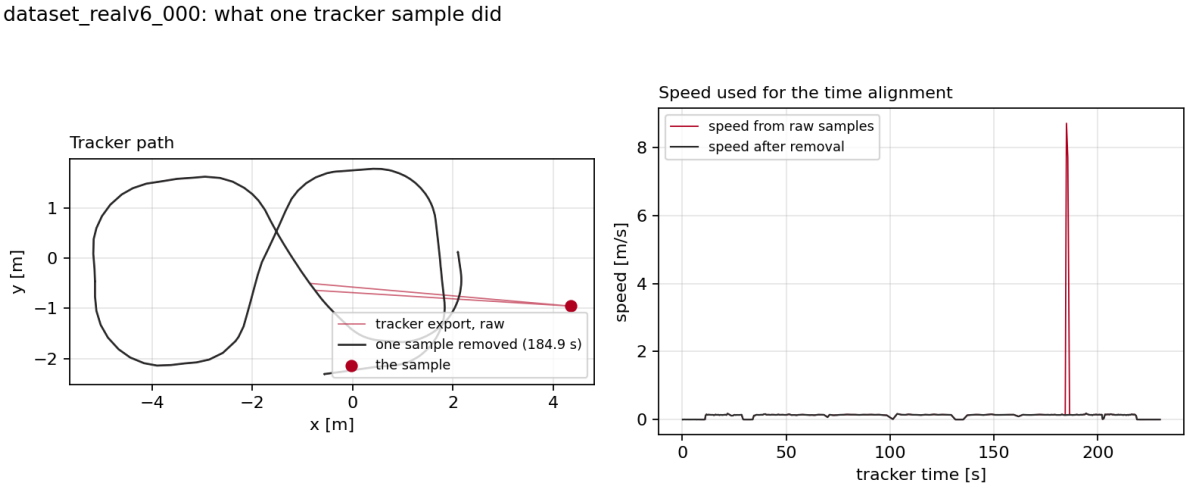


Figure I.5: The tracker export of `dataset_realv6_000`: one sample at 184.9 s lies 5.2 m off the path and back (left), which puts an 8.7 m s^{-1} spike into the speed profile the time alignment correlates against (right).

adds 10 m to the true path length (37.3 against 27.1 m) and dominates the speed cross-correlation; scored against it the GPU run reads 1.17 m with a path ratio of 0.65, and with that one sample removed, and nothing else, 0.52 m with a path ratio of 0.85 (table I.8). The sparse rate and the gaps remain, so the cleaned truth still fails the gap rule; the cleaning is recorded here as a one-sample edit with its effect shown, not applied silently.

Table I.9: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, 3 November 2026 afternoon series: repaired library, replay rate 1.0, 3 November stereo extrinsic, `IMU.NoiseGyro` 0.1 and `IMU.NoiseAcc` 1.0, GPU or CPU feature extractor, loop closing (LC) on or off. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational; path ratio = estimate / truth; resets = `local_mapping_bad_imu` map resets; dropped = camera frames the node's queue discarded of about 6 480 received. Lever arm tracker-marker to IMU unmeasured (zero applied); time offset pinned to −4.14 s; all values provisional under the scorer's validity rules.

Run	ATE [m]	median [m]	max [m]	RPE_{1s} [m]	path ratio	resets	dropped
ORB-SLAM3, GPU, LC on	0.52	0.44	1.11	0.057	0.85	2	2420
ORB-SLAM3, CPU, LC on	0.62	0.48	1.38	0.110	0.88	2	4363
ORB-SLAM3, GPU, LC off	0.46	0.36	0.99	0.056	0.87	0	3201
ORB-SLAM3, CPU, LC off	1.03	0.89	1.83	0.223	0.96	3	4204

ORB-SLAM3 tracks the whole route in all four configurations, and the front end is what separates them. The two GPU runs score 0.52 (loop closing on) and 0.46 m (off, with zero resets) after dropping 2 420 and 3 201 of about 6 480 frames; the two CPU runs drop 4 363 and 4 204 frames, two thirds of the stream, and score 0.62 and 1.03 m, the first with a 2 m vertical excursion the scorer flags as implausible and the second with three resets and a half-second of lost tracking (table I.9, figure I.6). Loop closing changes nothing that the front end does not change more. After the initialisation resets no run leaves the OK state; the error is a slow wave between 0.05 and 1 m, not an accumulating drift (figure I.7), and it does

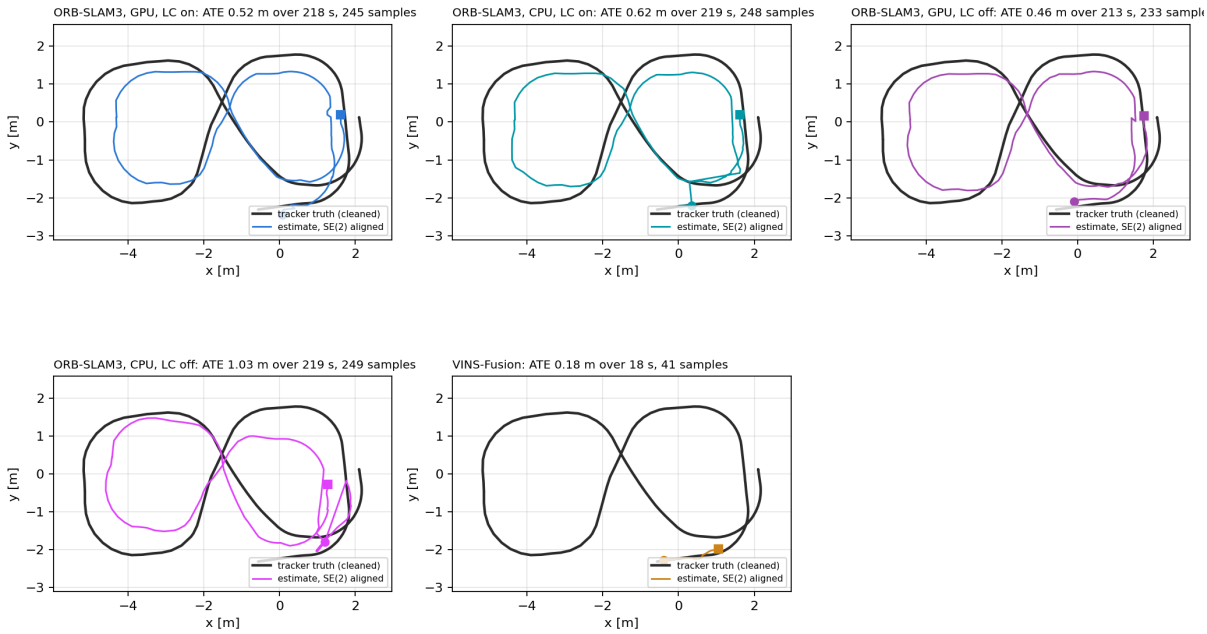


Figure I.6: Every surviving run against the cleaned tracker truth after rigid SE(2) alignment, one panel per run: ORB-SLAM3 with GPU or CPU extractor and loop closing on or off, and VINS-Fusion, whose 18 s before its resets took over are all it tracked.

Table I.10: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than $10\,^\circ\,\mathrm{s}^{-1}$ and on straights, with the time in each state. Positive = error growing. One run per configuration, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, GPU, LC on	-11.2	-0.3	23	171
ORB-SLAM3, CPU, LC on	-29.8	-1.4	23	162
ORB-SLAM3, GPU, LC off	4.9	-2.1	11	185
ORB-SLAM3, CPU, LC off	-49.9	-0.9	20	182

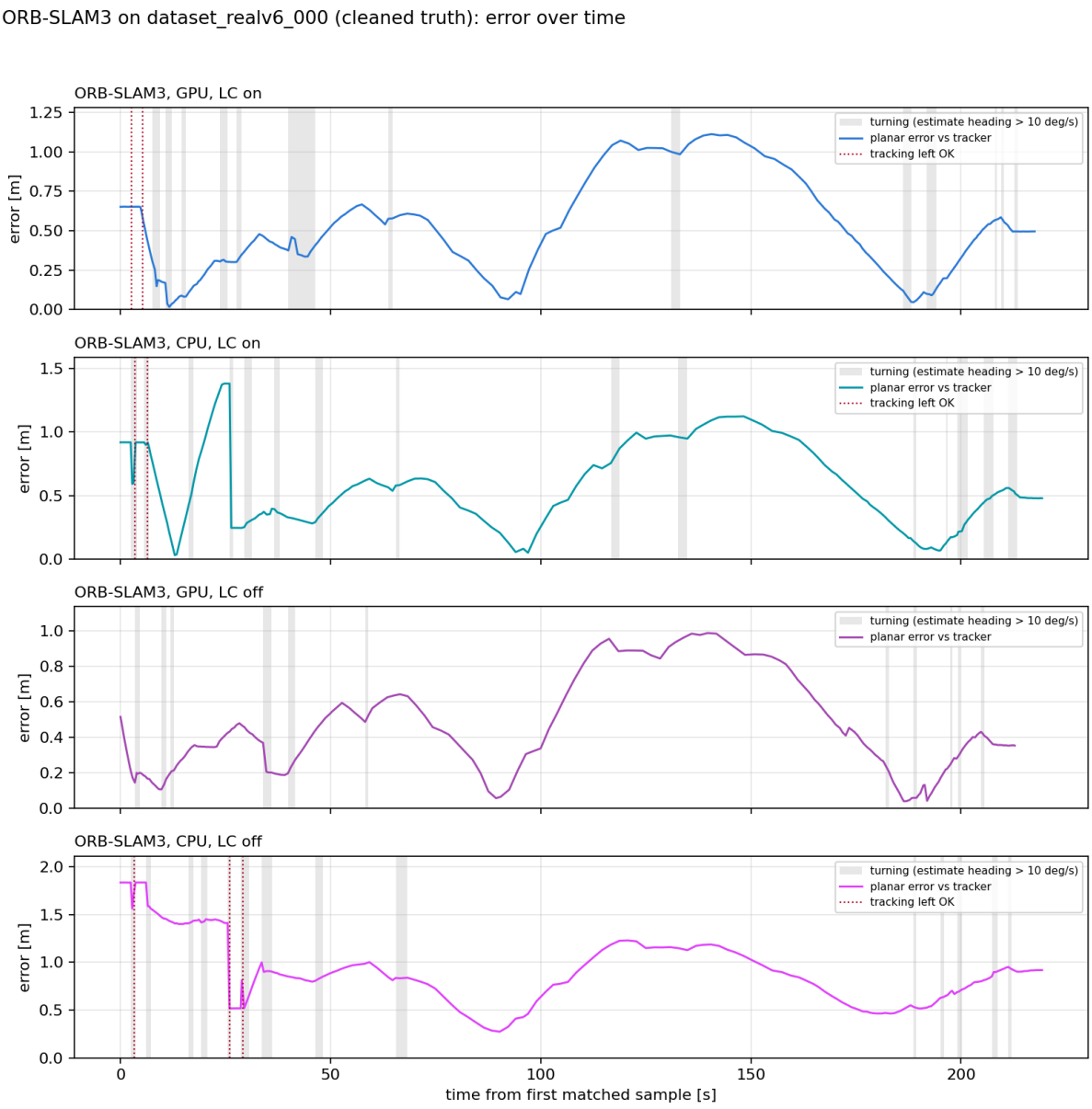


Figure I.7: ORB-SLAM3 planar error against the cleaned tracker truth over the run, one panel per configuration. Grey bands: the aligned estimate's heading changes faster than $10\,^\circ\,\mathrm{s}^{-1}$. Dotted red: the frames at which tracking left the OK state (the `local_mapping_bad_imu` initialisation resets, all inside the first 13 s).

not rise while turning in any of the four (table I.10), the same answer the simulation gave once the lever arm was removed.

What the 0.5 m is made of is not settled. The estimates' loops are a uniform 12–15 % smaller than the tracker's (path ratios 0.85–0.88 in the three plausible runs; figure I.6), which a stereo-baseline error, a tracker scale error or the unmeasured lever arm could each produce.

6.1 How the setting was reached

The configuration above was not the first tried. The runs of 2 November and of the morning of 3 November, whose directories these re-runs overwrote, showed that the stereo-inertial estimator with the IMU noise densities copied from the VINS file was unstable on this recording: a minority of runs stayed below 1 m and the rest ran away in the velocity state at a different place each time, while the robot crept at $0.12\,\mathrm{m\,s^{-1}}$. Six out-of-bounds reads in the stereo-fisheye code path were found and fixed, an AddressSanitizer run over the whole recording came out clean, and keyframe culling, loop closing, the GPU front end, the replay rate, frame drops, the stereo extrinsic and the initialisation were each varied without removing the runaway; raising the IMU noise densities tenfold removed it in every run since. The reading is that the inertial optimisation was trusting an IMU whose camera-relative axis signs, lever arm and time offset are unverified; with the weaker weighting the images carry the solution and the IMU acts as a prior, which is the role the evidence supports. The search is recorded in the ORB-SLAM3 engineering record (sections 7.10 to 7.13 and the status table); no number from those overwritten runs is quoted here.

Table I.11: VINS-Fusion on `dataset_realv6_000`, 3 November 2026 (`config/wil/stereo_imu.yaml`, IMU noise $\times 10$, gravity norm still 9.2642 against the $9.878\,\mathrm{m\,s^{-2}}$ this IMU reads at rest). The scored span is the 18 s it tracked before its last reset and is not a result.

Quantity	Value
camera frames received (bag: 6 601)	3187
IMU messages received / samples used	43494 / 6498
`imu_gap` resets	217
poses written / span of poses	286 / 19 s
tracker samples matched	41 of 43 in span
ATE over the tracked span	0.18 m (not a result)

VINS-Fusion has no valid run on this recording. The node received 3 187 of the bag's 6 601 frames but used only 6 498 of the 43 494 IMU messages it received; it raised 217 `imu_gap` resets and wrote 19 s of poses before the resets took over (table I.11, figure I.8). The delivery of the IMU stream into VINS-Fusion is therefore the defect in front of every other; behind it stands a gravity norm of 9.2642 in the configuration against the $9.878\,\mathrm{m\,s^{-2}}$ this bag's IMU reads at rest, which the overwritten 2 November runs had shown to drive a velocity runaway once the visual constraints thin. The 18 s it did track scored 0.18 m against 41 tracker samples, which is too short to mean anything and is not a result.

6.2 realv7: three recordings with a valid tracker truth, and an IMU initialisation that the map reset threw away

The three `dataset_realv7` recordings of 3 November 2026 (routes of 21.6, 31.7 and 23.4 m over 97, 134 and 104 s, a rounded rectangle, a figure-of-eight and a rounded rectangle) are the first whose tracker export passes every validity rule of the scorer: 5–6 Hz, no gap above 1 s, no step. On them the same ORB-SLAM3 setting that tracked realv6 produces an estimate that lies in a tilted plane in nine of twelve runs, and the tilt, not the configuration, decides the score. Tables I.12 and I.14 and figures I.9 and I.10 carry the observation as scored on 3 November. The tilt was read first as a wrong gravity direction and then as a frame fixed by the node; the per-frame diagnostics of the runs, read on 4 November, give the

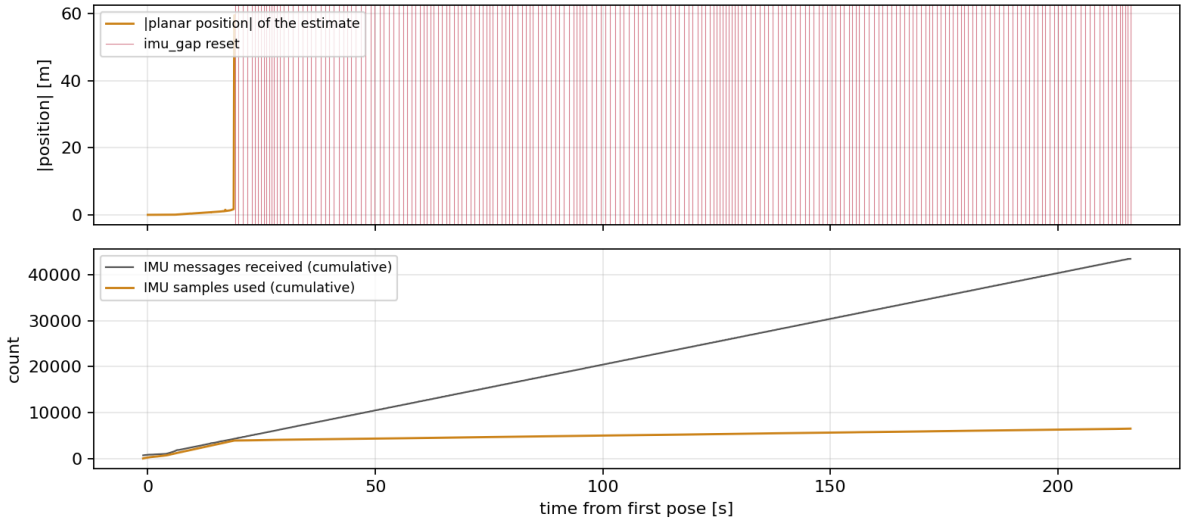


Figure I.8: VINS-Fusion on `dataset_realv6_000`: distance of the estimate from its origin with every `imu_gap` reset marked (top), and the IMU messages the node received against the samples the estimator used (bottom).

mechanism (below), and a repair in the scorer that takes nothing from the truth makes all twelve runs scorable (table I.13, figure I.11). The 3 November scoring of `001` and `002` took the time offset from the estimate's own speed because their bags were not on the machine; the repaired scoring uses the wheel speed on all three.

The tilt is real, and it is neither the gravity estimate nor the node: it is the inertial initialisation being thrown away. A plane fitted to each estimate's three-dimensional positions lies within 10° of horizontal in three of the twelve runs and at 88–90° in five, with the rest in between (table I.12); on `realv7_000` all four configurations give a vertical plane with the route intact in it (figure I.10, top row), and the normal of that plane is in every run exactly one of the body axes. The node writes ORB-SLAM3's world frame as it is (its first-pose alignment option is off), and the per-frame diagnostics of the runs tell one story in nine of the twelve. The IMU initialised 3–4 s after the first pose, at which moment ORB-SLAM3 rotated its map so that gravity points down z (the 0.38 m step in the log is that rotation applied to a robot 0.38 m from the origin). Within two seconds the upstream rule that resets a map whose consecutive keyframes lie closer than 2 cm in the ten seconds after the initialisation fired, because the robot was still standing or crawling, and the map was discarded. The new map starts in the camera frame of the reset instant, whose vertical is the camera's y axis, and in it the initialisation was starved: before the IMU is initialised a keyframe is inserted every 0.25 s, creating map points for one takes about 250 ms on this rig, and upstream attempts the initialisation only when its keyframe queue is empty, which it was for 3–9 % of the keyframes. The IMU re-initialised after 37–92 s in seven runs and never in two (`000` without loop closing and `001` with it, both GPU front end); only `001` with the CPU front end and loop closing re-initialised early, at 6.7 s, and it is the run that was level. Until the re-initialisation every pose is a stereo-only estimate in a map whose vertical is the camera's y axis, which is the body-axis normal; at the re-initialisation the map is rotated about its origin under the already-written poses, which is the 1.9–6.8 m step of the late-initialising runs. The IMU noise weighting that stopped the realv6 runaway and the IMU-to-camera rotation, whose axis signs remain unverified, play no part in this. The four surviving realv6 runs of table I.9 show the same reset at 3–5 s; their re-initialisation came after 9–34 s, which is why they were level.

Repaired, the twelve runs score (table I.13, figure I.11). The scorer gained two switches on 4 November, neither of which looks at the truth: it rotates the poses written before each initialisation by the rotation it measures at the initialisation frame, which brings the 0.38 and 6.8 m steps to 0.00 and

Table I.12: ORB-SLAM3 on the three `dataset_realv7` recordings, 3 November 2026 afternoon, same setting as the realv6 runs. "Tilt" is the angle between the plane fitted to the estimate's 3-D positions and the horizontal; "off-plane" the rms distance of the positions from that plane. A floor robot's estimate should be horizontal: a tilt near 90° means the poses were written while the map was not yet gravity-aligned (IMU not initialised: the vertical is then the camera's *y* axis), see table I.13 for the initialisation history and the repaired scores. ATE is planar after SE(2) alignment against the raw tracker export (which passes the validity rules on all three recordings); "offset" says how the time offset was found: *wheel* = wheel-speed cross-correlation with the bag present (realv7_000), *estimator* = the estimate's own speed with the bag absent (uncertain; superseded by the wheel offsets of table I.13). Lever arm unmeasured (zero).

Recording	Run	tilt [°]	off-plane [m]	z range [m]	ATE [m]	median [m]	path ratio	resets	offset
`realv7_000`	GPU, LC on	89.1	0.35	6.87	2.26	2.08	0.55	None	wheel
`realv7_000`	GPU, LC off	89.5	0.02	6.98	2.32	2.19	0.47	None	wheel
`realv7_000`	CPU, LC on	88.8	0.41	6.88	2.24	2.04	0.56	None	wheel
`realv7_000`	CPU, LC off	87.9	0.58	6.79	2.19	1.87	0.67	1	wheel
`realv7_001`	GPU, LC on	88.8	0.05	9.64	3.10	3.17	0.66	2	estimator
`realv7_001`	GPU, LC off	6.1	1.12	5.77	1.44	0.95	1.19	2	estimator
`realv7_001`	CPU, LC on	1.5	0.07	0.38	0.31	0.23	1.09	1	estimator
`realv7_001`	CPU, LC off	13.0	1.05	5.23	1.48	0.79	1.16	1	estimator
`realv7_002`	GPU, LC on	37.7	0.87	4.08	1.94	1.46	1.15	3	estimator
`realv7_002`	GPU, LC off	59.7	0.83	4.80	2.11	1.76	1.04	2	estimator
`realv7_002`	CPU, LC on	5.2	0.73	2.94	0.85	0.64	1.06	1	estimator
`realv7_002`	CPU, LC off	21.7	0.84	7.03	1.67	1.19	1.61	0	estimator

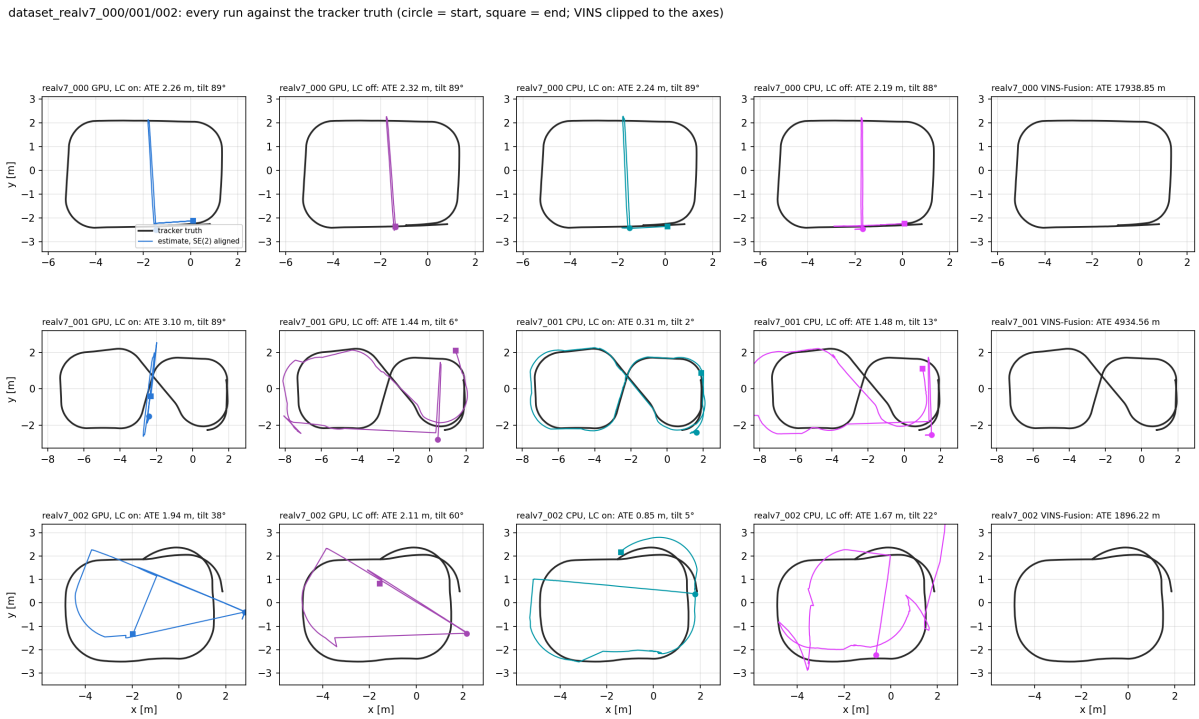


Figure I.9: Every `realv7` run against its tracker truth after rigid SE(2) alignment, one panel per run: three recordings (rows) by four ORB-SLAM3 configurations and VINS-Fusion (columns). Where the estimate's plane is vertical, the planar alignment sees the route collapsed onto a line.

0.03 m, and it levels each map's own fitted plane, the side chosen by the body's up axis, which agrees with the fitted normal within 2° wherever the plane is determined. On `000` the four configurations give 0.25–0.27 m planar ATE over 89 s, an RPE at 1 s of 0.05 m and path ratios of 1.05 to 1.09, with the IMU initialised for at most a quarter of the frames: stereo-only figures in all but name, and the best real-robot numbers of the project. On `001` the four give 0.44–0.82 m over 126 s; the run with the IMU initialised for 94 % of the frames scores 0.44 m, no better than the run with 3 %, and the 0.56 m run (CPU front end, loop closing off) keeps a vertical scatter of 1.02 m after levelling, two centimetres over the scorer's vertical-range rule, which labels it failed. On `002` one run tracks (0.47 m, CPU front end, loop closing), two lose tracking inside the route (resets at 49 and 59 s, 2.1 m ATE) and the fourth drifts without a reset (1.67 m, RPE 0.70 m, vertical range 4.3 m, failed by the same rule). The init-time resets of the other runs make them *partial* under the scorer's rules, as the table says. The remedy in the estimator is fork patch 15: two switches in the local mapper that attempt the initialisation while keyframes are queued and skip the insufficient-motion reset for stereo, whose scale is fixed. They are on in the real-rig configuration and untested on a recording; every number above was obtained without them.

Table I.13: ORB-SLAM3 on the three `dataset_realv7` recordings (3 November 2026, same setting as realv6), scored on 4 November with the bags on the dataset drive (wheel-speed time offsets) and the two frame repairs of `script/eval_real_gt.py`: the poses written before each IMU initialisation rotated into the frame used after it (`-join-imu-init`), and each map's fitted plane levelled (`-level-plane`). "IMU" is the share of tracked frames with the IMU initialised; "resets" the active-map resets after the first pose, in seconds (a reset restarts the position at the origin and is not stitched; those inside the route are tracking losses); "tilt" the plane tilt of the raw log, zero after the repair in every run. ATE is planar after SE(2) alignment, RPE over 1 s. Status from the scorer's rules ("partial" = a reset or loss the run recovered from). Lever arm unmeasured (zero); VINS-Fusion diverged on all three (table I.14) and is omitted.

Recording	Run	IMU [%]	resets [s]	tilt [°]	ATE [m]	median [m]	$RPE_{1\,s}$ [m]	path ratio	status
`realv7_000`	GPU, LC on	6	3	89	0.25	0.24	0.05	1.05	partial
`realv7_000`	GPU, LC off	0	3	89	0.26	0.21	0.05	1.09	partial
`realv7_000`	CPU, LC on	11	5	89	0.27	0.21	0.05	1.06	partial
`realv7_000`	CPU, LC off	23	4	88	0.26	0.23	0.05	1.08	partial
`realv7_001`	GPU, LC on	3	4	89	0.48	0.47	0.07	1.07	partial
`realv7_001`	GPU, LC off	67	3, 5	6	0.82	0.63	0.15	1.15	partial
`realv7_001`	CPU, LC on	94	4	2	0.44	0.40	0.07	1.07	partial
`realv7_001`	CPU, LC off	65	5	13	0.56	0.50	0.10	1.11	failed
`realv7_002`	GPU, LC on	22	3, 49, 59	38	2.11	1.28	0.26	1.18	partial
`realv7_002`	GPU, LC off	22	49	60	2.08	1.83	0.23	1.29	partial
`realv7_002`	CPU, LC on	61	none	5	0.47	0.41	0.12	1.13	partial
`realv7_002`	CPU, LC off	62	none	22	1.67	0.98	0.70	1.68	failed

Table I.14: VINS-Fusion on the three `dataset_realv7` recordings (`config/wil/stereo_imu.yaml`, IMU noise ×10, gravity norm 9.2642). Frames received against the bag's count (30 Hz over the recording), IMU use, resets, and where the estimate ended.

Recording	frames received	IMU received / used	resets	z range [m]	$\lvert p \rvert$ at end [m]
`realv7_000`	1155 of about 2663	17906 / 19053	0	12834403	37554
`realv7_001`	1541 of about 3686	24809 / 26340	0	3701	15974
`realv7_002`	1271 of about 2910	19368 / 20627	0	25937	5252

VINS-Fusion fails on all three in the manner of the first realv6 runs: the IMU stream now arrives without gaps (zero resets), 40–45 % of the frames reach the node, and the estimate diverges to kilometres on the stale gravity norm.

[What this slot still needs before it is a result] For realv6, a tracker export at its nominal 7 Hz without gaps, or an explicit rule for scoring on a sparse reference; for realv7, one run with the fork's two initialisation switches on, to show from the per-frame log that the IMU stays initialised from the first seconds without the scorer's repair; for both recording sets, the marker-to-IMU lever arm measured and

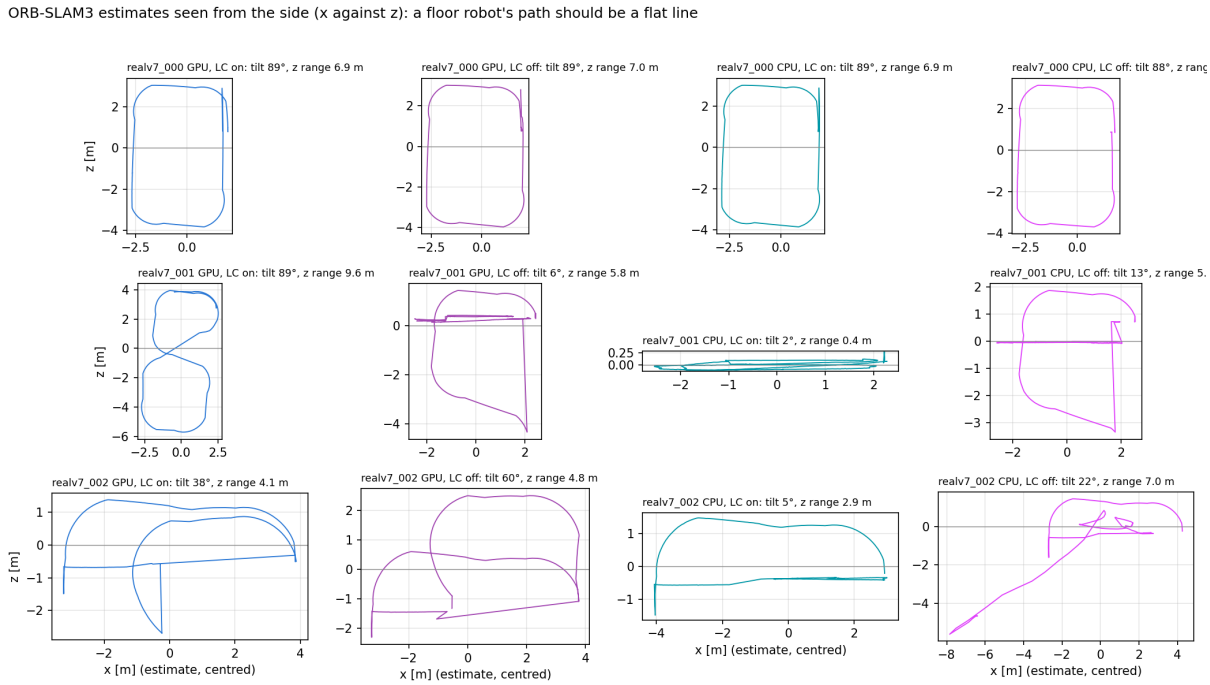


Figure I.10: The same twelve ORB-SLAM3 estimates seen from the side (estimate x against z, centred). A floor robot's path should be a flat line; on `realv7_000` every run is the route drawn in a vertical plane.

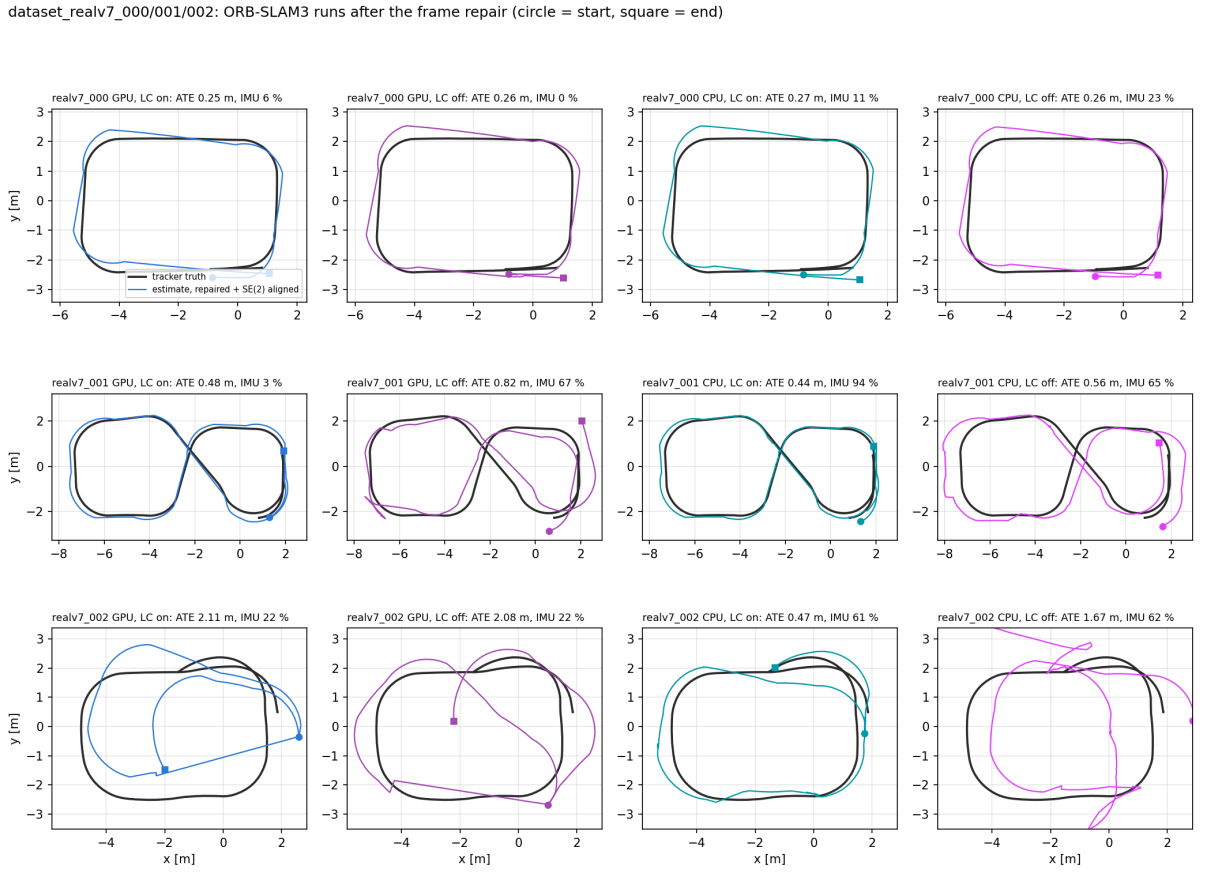


Figure I.11: The twelve `realv7` ORB-SLAM3 runs after the scorer's frame repair, against the tracker truth after rigid SE(2) alignment; each panel's title gives the planar ATE and the share of tracked frames with the IMU initialised.

the IMU-to-camera axis signs verified, which remain open although they no longer explain the tilt; a VINS-Fusion run whose node receives the IMU stream without gaps and the full frame stream, with `g_norm` set to what this IMU measures; repeats of the ORB-SLAM3 configurations kept under new names rather than overwritten; and at least one filtered run per estimator, without which Experiment 3's hypothesis has no real-robot test.

7 What these data cannot answer

- **Whether the filter helps.** No admissible filtered-versus-unfiltered pair exists in either simulated testbed, and no filtered run exists on the real robot; nothing here changes that.
- **Why VINS-Fusion fails in the extreme world.** The failure is immediate and not preceded by feature loss or by a rise in moving-object area; the mechanism is not identified.
- **The traffic effect with confidence.** Two to four unfiltered runs per estimator in the realistic world, with the people unmeasured, bound the effect rather than estimate it.
- **The feature-collapse signal, robustly.** The onset threshold is three times the static median error, and that median carries the lever arm. Recomputing the thresholds at the IMU point lowers them, moves the onsets earlier, and weakens the ORB-SLAM3+YOLO feature collapse of figure I.3 to a dip to about half in four runs (`fig_onset_epoch_lever0p338_light.png`). The collapse is real at the chapter's definition and threshold-dependent in size.

8 Summary

Consequences for Chapter 4. Re-score every simulation ATE with the ground truth moved to the IMU point before it is quoted, in Experiments 1 and 3. Report the extreme world as an immediate-failure testbed in which the filter cannot be evaluated, not as a drift testbed. Report the realistic world's traffic effect as bounded, not measured, and the class mask's feature cost as the one mechanism these data show. For the real-robot slot, quote ORB-SLAM3's 0.46–0.52 m (GPU front end) as provisional with the sparse-truth and lever-arm caveats attached, report the VINS-Fusion failure as a delivery defect (IMU-stream gaps) in front of a configuration defect (stale gravity norm) rather than an estimator result, and keep the slot *partial* until a filtered run and a full-rate truth exist; report realv7 as the recording set with valid truth, ORB-SLAM3 at 0.25–0.27 m on `000`, 0.44–0.82 m on `001` and 0.47 m on the one `002` run without a loss, after the scorer's frame repair and with each run's initialisation history beside the number; the IMU stayed uninitialised for most of nine runs, so these are largely stereo-only figures until the fork's initialisation switches are exercised.

Table I.15: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q5 lever arm	Yes: 0.34 m of forward offset is in every simulation ATE	minimum of the sweep at the SDF value in 30 runs, two estimators, two testbeds	high
Q1 turning	No, once the lever arm is removed	contrasts of tables I.3 and I.4; heading error flat	medium (small groups)
Q2 object in view, extreme	Not measurable within a bag; moving area does not modulate tracking loss or jumps	table I.6	medium
Q2 traffic in view, realistic	No effect on the unfiltered drift; the class mask loses features and gains drift while traffic is in view (suggestive)	table I.7, figure I.4	low (n)
Q3 features before onset	ORB-SLAM3+YOLO yes (collapse to one fifth); VINS-Fusion no	figure I.3	medium
Q4 state at onset	On the first straight, not turning; objects in view	table I.5	high
Real robot, ORB	Tracks the whole 218 s route in all four configurations; provisional ATE 0.46 and 0.52 m with the GPU extractor, 0.62 and 1.03 m with the CPU extractor (two thirds of frames dropped); no turning effect; 12–15 % scale shortfall unexplained	table I.9 and figure I.6	medium (provisional, sparse truth, one run each)